A MULTI-MODAL FRAMEWORK FOR QUANTIFYING NARRATOR RELIABILITY IN ʿILM AR-RIJAL USING SEMANTIC SENTIMENT ANALYSIS AND GRAPH NEURAL NETWORKS

ALAN SYAHMI BIN SANIH @ SANI

FACULTY OF COMPUTING AND INFORMATICS

UNIVERSITY MALAYSIA SABAH

2026

NAME **:** ALAN SYAHMI BIN SANIH @ SANI

MATRIC NUMBER **:** BI23110344

TITLE **:** A MULTI-MODAL FRAMEWORK FOR QUANTIFYING NARRATOR RELIABILITY IN ʿILM AR-RIJAL USING SEMANTIC SENTIMENT ANALYSIS AND GRAPH NEURAL NETWORKS

DEGREE **:** BACHELOR OF COMPUTER SCIENCE WITH HONOURS

(DATA SCIENCE)

VIVA'S DATE **:** DD SEPTEMBER 2027

CERTIFIED BY;

1. **SUPERVISOR** Signature

DR. KHALIFA CHEKIMA

_____

DECLARATION

I hereby declare that the material in this thesis is my own except for quotations, equations, summaries and references, which have been duly acknowledged.

30 APRIL 2026



ALAN SYAHMI BIN SANIH @ SANI
BI23110344

ACKNOWLEDGEMENT

"خَيْرُ النَّاسِ أَنْفَعُهُمْ لِلنَّاس"
"*The best of people are those who are most beneficial to others.*"
– Prophet Muhammad ﷺ

(At-Tabarani in Al-Muʿjam al-Awsat Hadith: 5787, graded *Ṣaḥīḥ* by Al-Albani)

All praise and gratitude belong to Allah for giving me the strength, patience, and clarity to complete this proposal. This work is a small attempt to serve the tradition of ʿIlm ar-Rijal, and I pray that it benefits others in some way.

I would like to express my deepest thanks to my supervisor, Dr. Khalifa Chekima, for his guidance, patience, and honest feedback throughout this journey. He gave me the freedom to explore an ambitious idea while quietly making sure I didn't go off the rails.

I am also grateful to the researchers and developers behind the OpenITI corpus, Shamela Digital Library, and CAMeL Tools, as their work made this project possible in the first place. Likewise, I thank the authors of the AR-Sanad 280K dataset and the Ifta' Sunnah platform for making high-quality Arabic Hadith data accessible to the research community.

To my parents and family: thank you for your endless prayers, for believing in me and for never asking "when will you finish?" with the wrong tone. This degree is as much yours as it is mine.

Finally, to my friends who listened to me ramble about narrators and *isnād* at odd hours, you know who you are. Thank you for the many "spinning" sessions and for the generous amount of ideas we all made.

May Allah accept this small effort of mine, forgive my shortcomings, and make it beneficial to those who study His religion.

Alan Syahmi Bin Sanih @ Sani
8 May 2026

ABSTRACT

The authentication of Prophetic reports (Ḥadīth) through the science of narrator criticism (*ʿIlm ar-Rijal/Jarḥ wa Taʿdīl*) is a cornerstone of Islamic scholarship, yet the digital era has outpaced manual verification methods. This project proposes a multi-modal framework that automates narrator reliability scoring by integrating three computational modalities: Arabic Named Entity Recognition (NER) using a fine-tuned AraBERT v2 model to extract narrator names, kunya, and nisba from classical biographical texts (Taqrīb at-Tahḏīb of Ibn Ḥajar al-ʿAsqalāni and the Tahḏīb al-Kamāl of al-Mizzī) and *isnād* chains (AR-Sanad 280K); a Knowledge Graph (KG) constructed with NetworkX (~1,500 narrator nodes, verbal edges e.g. heard_from and reported_from, rich attributes including *tabaqa*, travel history, and classical *Jarḥ wa Taʿdīl* labels) to enable structural and temporal integrity checks; and semantic sentiment quantisation that maps qualitative terms (e.g., *Thiqa*, *Ṣadūq*, *Kaḏḏāb*) to continuous reliability weights (0.00–1.00) and applies edge-level risk modifiers (e.g., 0.80 for *tadlīs*). The final chain score is computed as the product of node weights and edge modifiers, then mapped to *Ṣaḥīḥ/Ḥasan/Ḍaʿīf/Mawḍūʿ* grades. Evaluation against ground-truth gradings from the *Kutub as-Sitta* (the Six Canonical Hadith Collections; i.e. Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmiḏī, Sunan al-Nasāʾī and Sunan Ibn Mājah) uses accuracy, F1, RMSE, and Cohen's Kappa. The framework addresses the research gap of end-to-end automated trust scoring, providing an explainable, reproducible decision-support tool for Ḥadith scholars.

Keywords: Hadith authentication, ʿIlm ar-Rijal, Jarḥ wa Taʿdīl, Arabic NER, Knowledge Graph, Graph Neural Networks, semantic sentiment quantisation, multi-modal framework

ABSTRAK

RANGKA KERJA MULTIMOD UNTUK MENGUANTIFIKASI KEBOLEHPERCAYAAN PERAWI DALAM ILMU RIJAL MENGGUNAKAN ANALISIS SENTIMEN SEMANTIK DAN RANGKAIAN NEURAL GRAF

Pengesahan riwayat-riwayat Nabi Muhammad (hadis) melalui ilmu kritik perawi (ilmu rijal/al-jarh wa al-ta'dil) merupakan suatu tunjang kesarjanaan Islam, namun era digital telah mendahului kaedah-kaedah pengesahan manual. Projek ini mencadangkan rangka kerja multimodal yang mengautomasikan penskoran kebolehpercayaan perawi dengan mengintegrasikan tiga modaliti pengiraan: Pengecaman Entiti Nama (NER) Bahasa Arab dengan menggunakan model AraBERT v2 yang ditala halus untuk mengekstrak nama perawi, kunyah dan nisbah daripada teks biografi klasik (Taqrīb al-Tahḏīb karya Ibn Ḥajar al-ʿAsqalānī dan Tahḏīb al-Kamāl karya al-Mizzī) serta rangkaian isnad (AR-Sanad 280K); Graf Pengetahuan (KG) yang dibina dengan NetworkX (~1,500 nod perawi, tepi lisan seperti heard_from dan reported_from, atribut kaya seperti tabaqa, sejarah perjalanan, dan label Jarḥ wa Taʿdīl klasik) untuk membolehkan pemeriksaan integriti struktur dan temporal; dan kuantifikasi sentimen semantik yang memetakan istilah kualitatif (contohnya, Thiqa, Ṣadūq, Kaḏḏāb) kepada pemberat kebolehpercayaan berterusan (0.00–1.00) serta menggunakan pengubah suai tepi (contohnya, 0.80 untuk tadlīs). Skor rantai akhir dihitung sebagai hasil darab pemberat nod dan pengubah suai tepi, kemudian dipetakan kepada gred Ṣaḥīḥ/Ḥasan/Ḍaʿīf/Mawḍūʿ. Penilaian terhadap penggredan kebenaran daripada Kutub as-Sittah (Enam Kitab Hadis kanonik iaitu Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmiḏī, Sunan al-Nasāʾī dan Sunan Ibn Mājah) menggunakan ketepatan, F1, RMSE, dan Kappa Cohen. Rangka kerja ini menangani jurang penyelidikan dalam penskoran kebolehpercayaan automatik dari hujung ke hujung, langsung menyediakan alat sokongan keputusan yang boleh dijelaskan dan dihasilkan semula untuk ulama hadis.

Kata kunci: Pengesahan Hadis, Ilmu Rijal, Al-Jarh wa al-Ta'dil, NER Bahasa Arab, Graf Pengetahuan, Rangkaian Neural Graf, kuantifikasi sentimen semantik, rangka kerja multimodal

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CHAPTER 1

INTRODUCTION

1.1 Overview

The authentication of prophetic traditions (Sunna) through the reports attributed to the Prophet Muhammad (Hadith) forms one of the most intricate scholarly endeavours in classical Islamic civilisation. To achieve such rigorous requirements, the science of evaluating the men and women who form the chains of narration (*isnād*) through which a Hadith is transmitted is then employed. In Arabic, this science is known as *ʿIlm ar-Rijal*. Additionally, an inner layer of the science known as the discipline of *Jarḥ wa Taʿdīl* (literally "criticism and commendation") provides the epistemological framework by which scholars assess the moral probity (*ʿadālah*) and mnemonic precision (*ḍabṭ*) of every individual narrator in a given chain (Kamali, 2014). Over the centuries, this civilisational practice produced vast historical corpora of biographical literature, such as Ibn Hajar al-ʿAsqalāni's twelve-volume Tahdhīb at-Tahdhīb, representing thousands of qualitative evaluative judgements that collectively determine the trustworthiness of the prophetic record.

The digitisation of classical texts through global repositories in recent years has made the Hadith corpus accessible on an unprecedented scale. Yet paradoxically, this very accessibility has created a new challenge; the capacity of modern scholarship to rigorously authenticate and verify Hadith at the speed of digital dissemination has not kept pace with the volume of their circulation. Recent work by Mosa (2025) highlights a pervasive issue of general anxiety, wherein the rapidity of digital access frequently surpasses the capacity to authenticate transmission chains. The manual cross-referencing labour of biographical entries across hundreds of classical volumes remains an expert-dependent process that is resistant to scale, thus presenting a compounding challenge for the field as a whole (Mosa, 2025).

In hopes of closing such gaps, this project proposes to develop a multi-modal and data-driven framework that automates and hopefully streamlines the whole process of

narrator reliability evaluation while being grounded in the classical principles of Jarḥ wa Taʿdīl. The framework shall also aim to integrate these three advanced computational techniques:

1. Transformer-based Arabic Named Entity Recognition (NER) for the precise extraction and disambiguation of narrator identities from classical texts (Muther et al., 2023);
2. Graph Neural Networks (GNN) for modelling the structural and temporal integrity of *isnād* transmission chains (Mahmoud et al., 2024); and
3. Semantic Sentiment Quantisation to translate qualitative scholarly evaluations into a standardised, continuous reliability index (Ghriss, 2025).

The integration of these methodologies is significantly informed by foundational studies, including Mahmoud et al. (2022) on narrator disambiguation, Farooqi et al. (2024) on structured transmission datasets, and sequence-based *isnād* modelling by Najeeb (2022) and Oumer (2025). This research seeks to develop a reproducible, explicable, and epistemologically robust system that directly enhances the traditional discipline of *ʿIlm ar-Rijal* through contemporary data science techniques.

1.2 Problem Background

The evaluation of Hadith authenticity through *ʿIlm ar-Rijal* remains a cornerstone of Islamic scholarship, as the credibility of a prophetic tradition is fundamentally tied to the reliability of its chain of transmission (*isnād*). This rigorous verification system necessitates that every link in the chain be scrutinised for both moral integrity (*ʿadāla*) and mnemonic precision (*ḍabṭ*) (Kamali, 2022). Traditionally, this process is a labour-intensive endeavour that requires scholars to manually cross-reference thousands of qualitative biographical entries across vast historical corpuses, such as the 12-volume Tahdhib at-Tahdhib of Ibn Hajar or the foundational Andalusian Maliki assessments in Ibn ʿAbd al-Barr's At-Tamhid (Al-Asqalani, d. 852 AH; Ibn ʿAbd al-Barr, d. 463 AH).

Feasibility studies have confirmed that machine learning can indeed assist in automating *isnād* analysis, provided that the nuances of classical Arabic and the methodological standards of traditional jurisprudence are preserved (Usman, 2025; Mahmoud et al., 2022). The consonantal script and orthographic ambiguity of classical texts, however, make this particularly challenging, as minor variations in name orthography can lead to misidentification of narrators across manuscript traditions (El Ganadi et al., 2025). This results in an urgent need for Natural Language Processing (NLP) techniques to extract and

disambiguate narrator identities, and for Graph Neural Networks (GNN) to model the complex relational networks those narrators form (Farooqi et al., 2024; Mahmoud et al., 2022).

Recent work has confirmed the technical viability of this approach, where transformer-based models like AraBERT have achieved high accuracy in identifying ambiguous narrators from biographical text, while GNN frameworks can generate node embeddings to capture the relative position and credibility of a narrator within the broader transmission network (Ahmad et al., 2025; Mahmoud et al., 2022). By transitioning to a systematically quantified framework, researchers can achieve higher consistency and speed in evaluation without compromising the epistemological roots of the science (Tarmom, 2022).

1.3 Problem Statements

Two primary and significant challenges remain in the computational study of Hadith despite promising advancements in Arabic NLP:

i. While current models have improved in Named Entity Recognition (NER) for extracting narrator names and lineages, there is a critical lack of computational frameworks capable of quantifying narrator reliability. Identifying a narrator is insufficient, as the framework must be able to "judge" their standing based on historical evidence (Ahmad et al., 2025).
ii. Recent work by Mosa (2025) and Oumer (2025) has shown that transformer models are capable of extracting narration chains from Hadith text as graphs. Nonetheless, those approaches stop at the graph, as they did not assign a reliability score to any narrator or use any actual *Jarḥ wa Taʿdīl* praise or criticism terms to judge narrator trustworthiness. This research goes beyond graph construction by adding a quantitative reliability index and GNN-based structural analysis.

1.4 Project Objectives

The main objectives of this project are as follows,

i. To develop an NLP classification model that maps qualitative scholarly judgments to a standardised, quantitative reliability index (0.0 - 1.0).
ii. To construct a Knowledge Graph representing teacher-student relationships and utilise GNNs supported with geospatial data to evaluate the structural integrity and geographical feasibility of transmission chains.
iii. To evaluate the framework's accuracy by comparing automated reliability scores against established scholarly Hadith gradings (*Ṣaḥīḥ*, *Ḥasan*, *Ḍaʿīf*).

1.5 Project Scope

To ensure that the research remains focused, this project operates within a clearly defined set of boundaries, a technically feasible methodology, and is evaluable within the constraints of a single academic year.

1.5.1 Data Sources

The framework will be trained and tested exclusively on digitised classical Arabic biographical texts available through the Shamela digital library, specifically the Taqrīb at-Tahdhīb of Ibn Hajar al-Asqalāni and the Tahdhīb al-Kamāl of al-Mizzī. These texts contain the most

comprehensive and widely referenced collections of narrator biographical assessments and thus represent the most suitable ground-truth source for *Jarḥ wa Taʿdīl* labels.

1.5.2 Scale and Dataset

The initial experimental scope targets approximately 1,500 narrator biographical entries for building the reliability model and 1,000 validated Hadith *isnāds* drawn from the *Al-Kutub al-Sittah* (The Six Canonical Books, i.e. Ṣaḥīḥ al-Bukhari, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmiḏī (Sunan at-Tirmiḏī), Sunan al-Nasāʾī and Sunan Ibn Mājah) for end-to-end testing. While this is a small scale compared to the hundreds of thousands of available data points, this scale remains consistent with comparable feasibility studies in the literature (Mahmoud et al., 2022; Najeeb, 2022) and is sufficiently large to support statistically meaningful evaluation without exhausting available computational resources.

1.5.3 Technical Scope

The system will focus on four interconnected tasks,

i. Narrator extraction: Identifying proper names, kunya (honorifics), and nisba (place-based identifiers) from *isnād*;
ii. Entity disambiguation: Resolving alias variations to canonical narrator identities via a Knowledge Graph;
iii. Reliability scoring: Mapping aggregated *Jarḥ wa Taʿdīl* labels to a continuous index $W = [0.0 - 1.0]$. The primary programming language is Python, utilising Arabic-specific NLP libraries (e.g., CAMeL Tools, PyArabic) and transformer models (e.g., AraBERT, MARBERT);
iv. Geospatial Mapping: Extracting narrator locations (*nisba*) and mapping them to coordinates to verify if a student and teacher were in the same region at the same time.

1.5.4 Application Domain Limitations

This research is strictly concerned with the **structural evaluation of *isnād*** (*sanad* criticism). The analysis of Hadith textual content (*Matn* criticism) requires comparative jurisprudential reasoning across the four Sunni schools and falls outside the scope of this project. Furthermore, any automated reliability score produced by the framework is intended as a **decision-support tool** for scholars, not as a definitive Islamic ruling or fatwa. The epistemological authority of qualified human *muhaddithun* is expressly preserved.

1.5.5 Out of Scope

The following are explicitly excluded from this project:

a. *Matn* (textual body) analysis or content-based Hadith grading,
b. Coverage of Shīʿa *isnād* evaluation methodologies,
c. Real-time or web-based deployment of the scoring pipeline, and
d. Languages other than Arabic (e.g. Urdu or Persian Hadith commentaries).

CHAPTER 2

LITERATURE REVIEW

2.1 Hadith and its Structure

According to Almaany (n.d.), a Hadith (حَدِيث) is an Arabic noun that means a "report; account; tale; narrative; yarn" or a "talking; discourse; words; utterance; address; conversation; dialogue; saying; speaking; speech; talk". Islamically, Hadith due to its nature as the embodiment of the direct speech, talk, words, actions, approvals, and attributes from the Prophet Muhammad himself, plays a very important role in reconstructing the Sunna ("practice") as practiced by the Prophet and his companions (Shaffat, 2024). The Sunna on the other hand is essential in shaping the human understanding behind the divinely revealed law of the Sharia. If the Quran provides the broad legal axioms, the Sunna provides the operational parameters, and the Hadith serves as the transmission medium for those parameters, consequently making the Hadith corpora as the primary empirical data through which the Sunna is accessed and reconstructed (Nyazee, 2019).

In terms of structure, a Hadith consists of a *sanad* (chain of transmission, literally "support", plural: *isnād*) and a *matn* (content) (Shaffat, 2024). Below is an example of a Hadith and the aforementioned structure:

وَحَدَّثَنِي عَنْ مَالِكٍ، عَنِ ابْنِ شِهَابٍ، عَنْ عَلِيِّ بْنِ حُسَيْنِ بْنِ عَلِيِّ بْنِ أَبِي طَالِبٍ، أَنَّ رَسُولَ اللَّهِ صلى الله عليه وسلم قَالَ " مِنْ حُسْنِ إِسْلاَمِ الْمَرْءِ تَرْكُهُ مَا لاَ يَعْنِيه ".

And it was related to me (Yahya ibn Yahya al-Laythi) from Malik, from Ibn Shihab, from Ali ibn Husayn ibn Ali ibn Abi Talib, that the Messenger of Allah ﷺ said, "Part of the excellence of a man's Islam is that he leaves what does not concern him" (Mālik ibn Anas, 1989, Book 47, Hadith 3).

Based on the example Hadith, the part "And it was related to me (Yahya ibn Yahya al-Laythi) from Malik, from Ibn Shihab, from Ali ibn Husayn ibn Ali ibn Abi Talib, that the Messenger of Allah ﷺ said" can be considered as the *sanad* (chain of transmission), while the quote itself, "Part of the excellence of a man's Islam is that he leaves what does not concern him" is the *matn* (content). In order to grade Hadiths, the individuals specified in the *sanad* must have their reliability be verified, and this is done through the meticulous science of *ʿIlm ar-Rijal*, literally the "Studies of Men" or the more common Sunni and synonymous term, *Jarḥ wa Taʿdīl*, literally "critiscm and praise". Essentially, *ʿIlm ar-Rijal* or *Jarḥ wa Taʿdīl* functions as a metadata layer for the *sanad*, providing qualitative attributes such as reliability, memory, and lifespan for each node in the transmission graph (Indra Yuliasti et al., 2025; Ismail, 2024). A sanad therefore can be visualised as follows:

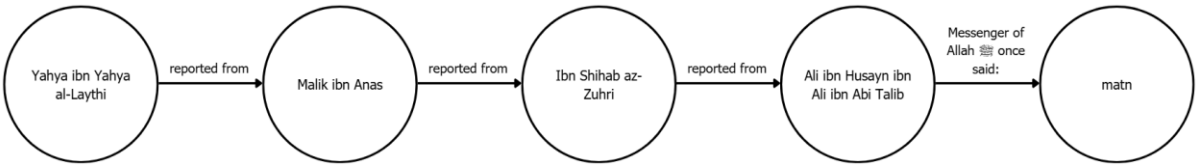


Figure 1.1: Graph Visualisation of a *Sanad*.

2.2 Impact of Hadith Grading

The grading of Hadith carries profound consequences for Islamic law (*fiqh*), theology (*ʿaqida*), and especially in everyday Muslim practice. Hadith that are classified as *Ṣaḥīḥ* (sound) or *Ḥasan* (good) are considered legally normative and form the second most authoritative source of Islamic jurisprudence after the Quran (Kamali, 2014). Conversely, acting upon *Ḍaʿīf* (weak) or *Mawḍūʿ* (fabricated) traditions without qualification risks introducing groundless practices into religious life (Umanah, 2024). The stakes of misclassification are therefore real and can be detrimental, as they directly affect rulings on worship, commerce, family law, and ethics observed by more than 1.9 billion Muslims worldwide.

The reliability of this grading system relies entirely on the truthfulness of the individuals who transmitted these reports across generations. A single unreliable narrator anywhere in the chain is sufficient to downgrade a Hadith, regardless of how credible the surrounding transmitters may be (Kamali, 2014). This sensitivity to individual narrator credibility implies that even small errors in evaluation, whether due to misidentification of a narrator or misapplication of a critical term, can disseminate into downstream jurisprudential decisions. Therefore, the cumulative impact of such errors, particularly in the rapidly expanding digital

Hadith landscape, represents a serious epistemological risk that this research directly seeks to mitigate (Umanah, 2024; Sri Wahyuningsih, 2025).

Looking beyond just the religious domain, the science of Hadith grading has also drawn increasing attention from historians, linguists, and information scientists as one of the oldest and most systematic peer-review and source-criticism methodologies in human intellectual history (Usman, 2025). The methodological rigour of classical *muhaddithun* (Hadith scholars), who developed chains of trust, biographical verification, and cross-referencing centuries before modern academic citation standards, offers a compelling analogue to contemporary knowledge verification challenges (Ghriss, 2025). Applying computational tools to this tradition, therefore, carries interdisciplinary significance that extends well beyond Islamic studies alone (A'thoina & Hasballah, 2025).

2.3 Hadith Authentication Process

The classical process of authenticating a Hadith is a dual-track methodology that evaluates both the *sanad* (chain of narrators) and the *matn* (textual body) of a tradition (Kamali, 2014). Sanad criticism, the focus of this research, concerns the continuity and credibility of the transmission chain from its originator (typically the Prophet Muhammad ﷺ) to the collector who recorded it. Each narrator in the chain must be proven to have met the preceding narrator, to have possessed an accurate memory, and to have been morally upright throughout their life. Any break in the chain (*inqita'*) or identified moral failing in a transmitter is sufficient grounds to weaken or reject the report entirely (Ahmed & Anas, 2025).

The grading system that emerges from this process is hierarchical and multi-tiered. At the apex sits the *Ṣaḥīḥ* ("sound") Hadith, which satisfies all criteria of chain continuity, narrator credibility, and absence of hidden defects (*ʿillah*) or contradictions (*shudhadh*). Below this are degrees of *Ḥasan* ("good"), where minor shortcomings in narrator memory are present but do not fatally undermine the report. *Ḍaʿīf* ("weak") Hadiths fail one or more of these conditions, whereas *Mawḍūʿ* ("fabricated") traditions are those for whose narrators deliberate falsehood has been established (Kamali, 2014). This taxonomy forms the classification target for the automated scoring system proposed in this research, where the aim is to computationally approximate the output of this traditional grading process (Ahmad et al., 2025).

The process of collecting the qualitative assessments that inform these grades is documented in a vast body of biographical literature known as *kutub ar-rijal* (books on narrators). Scholars such as Yahya ibn Maʿin, Ahmad ibn Hanbal, and later Ibn Hajar al-

ʿAsqalani dedicated their careers to interviewing, cross-referencing, and documenting the reliability status of thousands of narrators (Mghari et al., 2024; A'thoina & Hasballah, 2025). These books contain the raw *Jarḥ wa Taʿdīl* labels, i.e. terms of praise and criticism that this project aims to computationally extract, quantify, and aggregate into an automated reliability index.

2.4 ʿIlm ar-Rijal: An Expansive Biographical Study

ʿIlm ar-Rijal ("the science of narrators") or also synonymously known as *Jarḥ wa Taʿdīl* ("criticsm and praise") is one of the most elaborate biographical knowledge systems ever developed in any civilisation (Ahmed & Anas, 2025). At its peak, classical scholars tracked the life histories, teachers, students, travel records, and credibility assessments of over 500,000 narrators, a feat of multi-generational data collection that has no direct parallel in pre-modern scholarship (Usman, 2025). The core output of this science, the biographical dictionary (*muʿjam* or *tabaqat*), catalogues narrators by generation, geographic region, or alphabetical order, and assigns each an evaluative standing based on the consensus of prior critics (Maraoui et al., 2022).

The terminology employed within *ʿIlm ar-Rijal* is highly technical and hierarchically structured. Terms of commendation (*taʿdil*) range from the superlative *Thiqa Thiqa* (supremely reliable) down to *Saduq* (truthful but occasionally erring), with each grade carrying different implications for the admissibility of a narratorʿs reports. Terms of criticism (*jarh*) range from mild reservations such as *Layyin* ("soft", i.e., slightly weak) to severe condemnations such as *Kadhdhab* ("liar") or even *Dajjal* ("arch-deceiver") (Kamali, 2014). The precise semantic weight of each term and the conditions under which it strengthens or overrides competing assessments are the subject of a dedicated sub-discipline (Mghari et al., 2024), one that this project attempts to operationalise through Semantic Sentiment Quantisation.

A central computational challenge in this science is the problem of resolving **narrator identity**. The Arabic naming convention comprises a given name (ism), teknonym (*kunya*), patronym, matronym or a series of it (*nasab*), tribal affiliation (*nisba*), and nickname (*laqab*), which means that a single narrator may be referred to by dozens of orthographic variants across different texts and centuries (Mahmoud et al., 2022). The scholar known canonically as Abu Dawud as-Sijistani, for example, may appear across manuscripts in numerous abbreviated or dialectally inflected forms. Disambiguating these variants into a single canonical identity is a prerequisite for any reliable automated grading system, and is the primary

challenge that the Knowledge Graph and NER components of this research are designed to solve (Najeeb, 2022; Mosa, 2025).

2.5 Advancements in Arabic Named Entity Recognition (NER)

Modern processing of classical Arabic text has moved decisively away from rule-based constraints toward deep bidirectional representations. Research by Mahmoud et al. (2022) using the AR-Sanad 280K dataset established a new baseline for narrator disambiguation, proving that large-scale datasets are essential for handling the morphological density of biographical entries.

The current "State of the Art" (SOTA) by Mosa (2025) and Oumer (2025) utilises Pretrained Language Models (PLMs). Alghamdi et al. (2025) demonstrated that Transformer-based models like AraBERT and MARBERT significantly outperform previous iterations in identifying fraudulent content and extracting technical entities. These advancements facilitate high-accuracy Entity Linking, allowing the system to resolve identities across centuries even when names are identical (Mahmoud et al., 2022; Najeeb, 2022).

2.6 Knowledge Graphs (KG) for Entity Disambiguation

A Knowledge Graph (KG) is a structured representation of real-world entities and the semantic relationships between them, encoded as a directed graph of nodes and edges (Ji et al., 2022). In the context of *ʿIlm ar-Rijal*, this structure is particularly well-suited to the problem of narrator identity resolution: each node represents a canonical narrator identity, edges encode transmission relationships (teacher-student pairs), and node attributes store the accumulated biographical metadata, including *Jarḥ wa Taʿdīl* assessments drawn from classical *kutub ar-rijal* (Maraoui et al., 2022; Shafie, 2021). Unlike flat relational databases, KGs allow queries that traverse multi-hop relationships, enabling the system to infer that two orthographically different name mentions in separate manuscripts refer to the same historical individual based on their shared network neighbourhood (Hogan et al., 2021).

Within the domain of digital Hadith studies, several prominent Knowledge Graphs (KGs) have been developed to model the complex interdependencies of *isnād* chains (Alam, 2021). Systems such as **NarratorsKG** encode the topological structure of narrator transmission networks, mapping tens of thousands of distinct identities alongside their diverse *kunya* and *nisba* variants (Mahmoud et al., 2024). In a typical hybrid retrieval pipeline, an NER model extracts name mentions, which are then resolved against the KG's candidate pool using both local semantic context and global graph signals (Mahmoud et al., 2024).

Specifically, graph-level metrics like **PageRank centrality** are utilised to weigh a narrator's historical prominence; this helps resolve ambiguities where purely lexical matching fails, a common issue in the classical corpus where a single name may refer to five or more distinct individuals (Mosa, 2025).

More recently, the integration of KGs with Transformer-based models has defined the current state of the art. Mosa (2025) proposed a KG-Transformer framework that encodes narrator node embeddings directly into the attention mechanism of a pre-trained Arabic language model, allowing the model to simultaneously process the local textual context of a name mention and its global relational position within the transmission network. This synergy between structural and semantic representations addresses what earlier approaches treated as separate problems, which are text understanding and graph reasoning, and produces entity disambiguation accuracy that approaches or exceeds 97% on the AR-Sanad 280K-v2 benchmark (Mahmoud et al., 2024; Mosa, 2025). The construction and querying of such a KG therefore constitute a foundational prerequisite for the GNN-based reliability analysis described in the following section.

2.7 Advancements in Graph Neural Networks (GNN) for *Isnād* Analysis

Traditionally, the "structural integrity" of an *isnād* was analysed using Social Network Analysis (SNA), focusing on explicit metrics like degree centrality and betweenness to visualise transmission flows (Shafie, 2026; Siddiqui et al., 2023). While useful for visualisation, SNA is limited in its ability to learn latent relational features (Apgar et al., 2023).

The field has recently evolved toward Graph Neural Networks (GNN) and Knowledge Graphs (KG). Mahmoud et al. (2024) introduced narrator identification by utilising graph query embeddings and a knowledge graph (NarratorsKG), while Mosa (2025) proposed a Knowledge Graph-Transformer framework to synergise structure and semantics. Unlike SNA, GNNs can represent a narrator's "reliability" as a latent feature influenced by their entire neighbourhood of teachers and students. This allows for the detection of "hidden" patterns, such as temporal inconsistencies where a student claims to have heard from a teacher they could not have met, thus automating a core task of the *muhaddithun*.

A recent master's thesis (Oumer, 2025) successfully fine-tuned AraT5 and CAMeLBERT-CA to turn Hadith text into relational tuples and graph representations. While it is a useful structural work, it explicitly leaves out narrator reliability scoring, entity linking, and any quantification of *Jarḥ wa Taʿdīl* terms. This project aims to take graphs from the work as

a starting point and add the missing "trust layer", something clearly stated as out of scope in the aforementioned thesis.

2.8 Advancements in Semantic Sentiment Quantisation

A major challenge in automating Hadith grading is translating qualitative scholarly praise (*Ta'dil*) or criticism (*Jarh*) into numerical formats. Recent feasibility studies by Ahmad et al. (2024) confirm that Machine Learning is increasingly capable of automating *isnād* analysis when technical labels are correctly weighted.

Current advancements focus on Semantic Sentiment Quantisation, where technical terms such as *Thiqa* (reliable), *Saduq* (truthful), and *Munkar* (rejected) are mapped to specific reliability weights. Mghari et al. (2024) introduced Narrator2Vec, which uses word embeddings to create dense representations of a narrator's standing. This allows for a continuous reliability index $W \in [0,1]$, where the framework calculates a composite probability score for an entire chain by aggregating the individual quantised weights of each narrator. This move toward "Explainable AI" ensures that the quantitative output remains rooted in traditional Islamic epistemology (I'thoina & Hasballah, 2024). The synthesis of these three advancements, i.e. transformer-based identity resolution, GNN-driven structural analysis, and sentiment-based weight quantisation, forms the multi-modal foundation of this research.

2.9 The Research Gap: Automated Trust & Reliability Scoring

Despite the significant advancements documented in the preceding sections, a critical gap persists at the intersection of these research streams: no published study has yet delivered a complete, end-to-end pipeline that unifies narrator extraction, identity disambiguation, and probabilistic reliability scoring into a single, evaluatable system. The majority of existing work ceases at either the extraction or disambiguation stage, treating the production of a quantified trustworthiness score as a future direction rather than an achieved outcome (Ahmad et al., 2025). While text-based automated classification methods do exist (Tarmom, 2022), what remains absent is an end-to-end system that unifies knowledge graphs with probabilistic reliability scoring.

The separation between textual NLP work and graph-structural work further compounds this gap. Studies employing AraBERT or similar transformer models for NER do not, as a rule, feed their outputs into Knowledge Graph or GNN frameworks for structural validation (Alghamdi et al., 2025). Conversely, graph-based analyses of *isnād* networks rarely incorporate the semantic richness of biographical *Jarḥ wa Taʿdīl* texts as node features (Mosa,

2025; Mahmoud et al., 2024). This disciplinary fragmentation means that existing systems capture either the semantic dimension of narrator reliability or the structural dimension, but not both simultaneously. The multi-modal integration that would make automated grading genuinely robust remains a largely unexplored frontier (Ghriss, 2025).

The standardisation of evaluation metrics also presents a significant gap. While individual studies report precision, recall, or F1-scores for their specific sub-tasks, no shared benchmark protocol exists for assessing the accuracy of an end-to-end automated grading system against the ground-truth classifications found in the *Kutub as-Sitta* (Farooqi et al., 2024). This absence of a common evaluation framework makes it difficult to compare approaches across studies. Establishing such a benchmark protocol is therefore a secondary but also an important contribution of the present research (Mahmoud et al., 2022).

The closest existing work by Oumer (2025) builds a pipeline from Hadith text to a graph of narrators. While it is a real contribution, it does not answer the question that ʻIlm ar-Rijal actually asks: given this chain of narrators, how reliable are they? The output in the aforementioned work is a graph, while the aimed output of this project is a graded Hadith (*Ṣaḥīḥ*/*Ḥasan*/*Ḍaʿīf*) with an explainable score. No published study has yet combined semantic sentiment from *Jarḥ wa Taʿdīl* texts with GNN-based structural analysis to produce a narrator reliability score.

This project directly targets these gaps by proposing a unified pipeline that integrates transformer-based NER, Knowledge Graph entity linking, GNN structural analysis, and Semantic Sentiment Quantisation into a single framework evaluated against pre-graded Hadith from the canonical collections. This in turn advances the field from the current state of fragmented sub-system studies toward a reproducible and multi-modal solution grounded in both classical Islamic epistemology and contemporary machine learning practice.

CHAPTER 3

PROJECT METHODOLOGY

3.1 Research Design

This project adopts an experimental research design underpinned by the **Design Science Research Methodology** (DSRM). The central artefact is an end-to-end automated pipeline that ingests classical Arabic biographical text, extracts and disambiguates narrator entities, and produces a continuous reliability score for each narrator, thereby operationalising the traditional science of *Jarḥ wa Taʿdīl* through computational means. Development proceeds through three sequential stages, each corresponding to a distinct research objective defined in Section 1.4. Each stage produces a testable intermediate output that feeds into the next, ensuring that failures can be isolated and addressed incrementally.

The model will operate according to these three stages, 1) establishing the data foundation; 2) constructing the extraction and disambiguation engine; and 3) implementing the scoring model and end-to-end evaluation. This framework will add a probabilistic trust score for each narrator and use graph neural networks to evaluate chain-level reliability, defining the novelty of this project. Together, these phases form the technical pipeline as illustrated in Section 3.5 below.

3.2 Stage 1: Data Ingestion and Preprocessing

3.2.1 Data Sources

The framework ingests data from two primary streams to satisfy both the semantic evaluation of narrators and the structural evaluation of transmission networks:

a) Biographical Corpora: Digitized classical Arabic biographical evaluation texts (*kutub ar-rijāl*) are sourced directly from the Shamela Digital Library. The primary targeted texts are the Taqrīb at-Tahdhīb by Ibn Ḥajar al-ʿAsqalānī and the Tahdhīb al-Kamāl by al-Mizzī. These texts provide the critical *Jarḥ wa Taʿdīl* labels required to generate the foundational narrator reliability ground-truth index.

b) Transmission Chain Corpora: To train and evaluate the Named Entity Recognition (NER) boundary parsing modules, the system ingests the Sanadset 650K dataset, which provides 650,986 Hadiths with their respective *isnād* and book references. Additionally, an evaluation benchmark consisting of 1,000 validated *isnād* is systematically drawn from the hadith_datasets (obtained from HuggingFace) which contains the important 'Grade' column (e.g., "Sahih (Darussalam)").
c) Used exclusively for ground truth evaluation to serve as an end-to-end testing matrix.

3.2.2 Biographical Text Parsing and Block Extraction

The raw text documents acquired from the OpenITI corpus contain extensive architectural metadata, including custom YAML headers, page number selectors, and editorial comments that present noise to machine learning classifiers.

To isolate individual biographical entries (*tarājim*), a specialised Regular Expression (Regex) parser is deployed. The algorithm scans for historical entry identifiers and structural patterns typical of OpenITI formatting (e.g., `# $Tarjama` tags or numeric indices followed by narrator names). Once identified, the algorithm isolates each biography as an independent text block, discarding peripheral text, page numbers, and editorial notes.

3.2.3 Two-Tiered Algorithmic Alignment Framework

Because classical Hadith collections lack a universal indexing convention, records cannot be integrated using flat database joins on numerical indices. To cross-reference records accurately and efficiently, this project implements a reproducible, two-tiered hybrid alignment pipeline that combines sparse vector space modeling with token-level edit distance calculations. This architecture drastically reduces the search space, preventing computational bottlenecks while maintaining strict deterministic reproducibility.

a. Tier 1: Heuristic Search Space Reduction (Information Retrieval Layer)
    i. To avoid computing the Levenshtein distance across a computationally prohibitive $1{,}200 \times 650{,}000$ string pairs, a sparse vector retrieval layer is introduced to reduce the search space from the entire population ($M = 650{,}986$) to a narrow candidate pool ($K = 10$).
    ii. The normalised *Matn* strings of the entire Sanadset 650K corpus are fitted into a sparse TF-IDF (Term Frequency-Inverse Document Frequency) vectorizer using character n-grams ($n \in [3,5]$) to capture sub-word variations and typos.

iii. A stratified sample of 1,200 Hadiths is drawn from the hadith_datasets HuggingFace corpus, consisting of 200 Hadiths per canonical book from the Kutub as-Sitta (Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmiḏī, Sunan al-Nasāʾī, and Sunan Ibn Mājah). This per-book stratification ensures balanced representation across all six collections, providing sufficient statistical power (margin of error ≈ 2.8%) while remaining feasible within the project timeline. Each of the 1,200 sampled Hadiths from the HuggingFace corpus is tokenized and transformed into the same vector space.

iv. A cosine similarity dot product is calculated between the sample vector and the corpus matrix, extracting the top $K = 10$ highest-scoring candidate documents for each sample Hadith within milliseconds.

b. Tier 2: Fine-Grained Lexical Verification (Fuzzy Matching Layer)

i. Once the search space is isolated to the top 10 candidates per sample, a localized Levenshtein distance metric is applied via the rapidfuzz library to determine the absolute match:

$$\text{Similarity Ratio}(X, Y) = \frac{|X| + |Y| - \text{LevenshteinDistance}(X, Y)}{|X| + |Y|} \times 100$$

ii. A strict similarity threshold of ≥ 95 is enforced. In instances where multiple candidates yield identical fuzzy similarity ratios due to brief text lengths or overlapping phrasing, a secondary deterministic validation sweep is triggered.

iii. This module parses and cross-references the first two ancestral nodes (the immediate transmitters closest to the collector) within the Sanad sequence to break the tie.

c. Reproducibility and Quality Control Constraints

To guarantee that this pipeline can be fully replicated by future researchers, three strict compliance mechanisms are built into the source code:

i. Deterministic Seeding: The stratified sampling module fixes a global pseudo-random number of generator seed ($random_{state} = 42$), ensuring that the identical subset of 1,200 Hadiths is generated on every execution.

ii. Immutable Hashing: Upon a successful match, the records are merged and assigned an unalterable primary key (id) generated via a SHA-256 hash of the normalized consonantal text skeleton.

iii. Quarantine Routing Vector: Any record where the top candidate falls below the 95% fuzzy threshold or yields irreconcilable node mismatches during secondary

validation is automatically routed to an isolated JSON quarantine log for manual scholar-in-the-loop remediation, preserving the integrity of the primary training dataset.

3.3 Stage 2: Named Entity Recognition and Knowledge Graph Construction

3.3.1 NER Model Architecture

Narrator extraction is performed using a fine-tuned Transformer-based model. The base model is **AraBERT v2**, a pre-trained bidirectional encoder trained on a large Modern Standard Arabic corpus (Mahmoud et al., 2022). The model is fine-tuned on the Sanadset 650K dataset for a token classification task using the BIO (Beginning, Inside, Outside) tagging scheme. The following entity types are annotated:

Table 1: Example of Narrator Entity for the Companion Abu Hurayra

Tag	Entity Type	Example (Arabic)	Transliteration
B-NARRATOR	Beginning of narrator name	عبد ٱلرحمن	ʿAbd ar-Raḥmān
I-NARRATOR	Inside narrator name	بن صخر	bin Ṣakhr
B-KUNYA	Beginning of the teknonymic honourific	أبو	Abū
I-KUNYA	Inside the teknonymic honourific	هريرة	Hurayra
B-NISBA	Beginning of geographic/tribal affiliation	الدوسي	Ad-Dawsī
I-NISBA	Inside geographic/tribal affiliation	الزهراني	Az-Zahrānī
O	Outside any entity	حدثنا	"We narrated"

The model is trained for 10 epochs with a learning rate of 2^{e-5}, using the AdamW optimiser with a linear warmup schedule over the first 10% of training steps. Evaluation is performed on a held-out 20% test split of the Sanadset 650K dataset, with **micro-averaged F1-score** as the primary metric.

3.3.2 Knowledge Graph Construction

Extracted entity mentions are linked to canonical narrator identities stored in a **Knowledge Graph (KG)**. The KG is constructed using the Python 'NetworkX' library (with optional migration to Neo4j for scalability) and populated from the biographical entries in the Tahdhīb al-Kamāl. Each node in the KG represents a unique narrator and stores the following attributes:

Table 2: Attributes in a Rich Narrator Node

Node Attribute	**Description/Purpose**
ID	For entity disambiguation.
Name	The canonical name of the narrator.
Nasab	Patronym or chain of lineage.
Kunya	Teknonym (e.g., Abu Muhammad).
Nisba	Attributions based on tribe, place, or trade.
Common Name	The most famous or commonly used name when referring to the entity. May help entity disambiguation.
Birth/Death Dates	Dates for temporal integrity checks (where available).
Tabaqa	Generational marker to prevent anachronisms.
Ethnic/Legal Status	Helps determine if the transmission is localised/regional or widely accepted, and aids entity disambiguation.
Teacher/Student relationships (with duration)	Acts as the directed edges between nodes.
Travelled locations	To detect any inconsistencies or contradictions.
Recension (where available)	Boosts reliability if known to produce a critical revision of a document, such as the *Muwatta' Malik*.
Reliability	*Jarḥ wa Ta'dīl* labels are divided into inferred (machine-made conclusion) and classical (ground label truth).



Figure 2.1: Example of a Rich Narrator Node

Edges represent **transmission relationships** (teacher → student) extracted from *isnād* chains. When the NER model produces a name mention, the KG is queried using a combination of string similarity (Levenshtein distance on the consonantal skeleton) and graph neighbourhood matching (shared teachers/students) to resolve the mention to its most probable canonical identity.

To prevent any anachronisms and further augment the reliability of this system, an extended metadata system of *Tabaqat* (generations) based on Ibn Hajar's 12-tiered generational system will be implemented on every narrator. The generational system will cover from 622 CE (the start of the Islamic calendar) until 870 CE (around the mid-3rd century AH), and the narrators will be mapped to these discrete categories:

1. The *Ṣaḥāba* (Companions of the Prophet).
2. Senior *Tābiʿūn* (senior Followers of the Companions).
3. Intermediate *Tābiʿūn* (middle Followers of the Companions).
4. Junior *Tābiʿūn* (junior Followers of the Companions).
5. Senior *Atbāʿ at-Tābiʿīn* (senior students of the *Tābiʿūn*).
6. Intermediate *Atbāʿ at-Tābiʿīn* (middle students of the *Tābiʿūn*).

7. Junior *Atbāʿ at-Tābiʿīn* (junior students of the *Tābiʿūn*).
8. Senior *Kibar atbāʿ al-atbāʿ* (senior followers of the followers).
9. Intermediate *Kibar atbāʿ al-atbāʿ*.
10. Junior *Kibar atbāʿ al-atbāʿ*.
11. Later narrators (students of the great successors, up to the early compilers).
12. Compilers and their immediate students (ending around the late 3rd century AH / ~900 CE).

3.3.3 Expected Output

The output of Stage 2 is a populated Knowledge Graph containing approximately 1,500 canonical narrator nodes with linked aliases, biographical metadata, and transmission edges, alongside an NER model capable of extracting narrator mentions from unseen *isnād* with a target F1-score of ≥ 0.90.

3.3.4 Enriched *Sanad* Knowledge Graph

Each narrator node stores the full set of attributes listed in Table 1.2 (Section 3.3.2), including birth/death dates, travel history, recension, and inferred reliability score. For visual clarity, only ID, common name, tabaqa, and reliability are shown here.



Figure 2.2: Enriched *Sanad* Knowledge Graph Mock-up

3.4 Stage 3: Probabilistic Trust Scoring and Evaluation

3.4.1 Semantic Sentiment Analysis for Continuous Reliability Scoring

The core innovation of this stage is the translation of qualitative *Jarḥ wa Taʿdīl* labels into a continuous reliability score on the interval [0.0, 1.0], initially developed by Ghriss (2025) in their Iterative Co-Update Algorithm (ICUA) framework. This framework specifically formalises narrator reliability scores (R_N) on a scale of 0 to 1. This is achieved through **semantic sentiment analysis**, an NLP-based approach where it learns from the full contextual wording of scholarly assessments rather than relying on keyword-to-weight lookup alone. This proceeds in two parallel tracks that are compared empirically.

3.4.1.1 Track A: Transformer-Based Semantic Sentiment Classifier

A fine-tuned Arabic transformer model (AraBERT v2) is trained as a regression model to directly predict a reliability score from the raw text of a *Jarḥ wa Taʿdīl* statement. The training data consists of evaluative sentences extracted from the biographical entries in *Taqrīb at-Tahdhīb* and *Tahdhīb al-Kamāl*, where each statement is paired with its corresponding ordinal grade in the classical hierarchy (e.g., *Thiqa Thiqa*, *Thiqa*, *Ṣadūq*, *Layyin*, *Kaḏḏāb*). The model architecture is as follows:

- **Base model**: AraBERT v2, a bidirectional transformer pre-trained on ~70B tokens of Modern Standard and Classical Arabic text.
- **Classification head**: A pooling layer (mean of all token embeddings) followed by a linear layer with a single output neuron and a sigmoid activation, producing a score in [0.0, 1.0].
- **Loss function**: Mean Squared Error (MSE) between the predicted score and the normalised ground-truth grade.

Unlike a static lookup table, the transformer model captures **semantic context**: it learns that *"kāna thiqa ḍabt ḥāfiẓ"* ("he was reliable, firm, and had a strong memory") carries the same weight as the single term *thiqa* despite different wording. It can also detect when a positive term is negated (e.g., *"laysa bi-thiqa"*, "he is not reliable"), which a keyword lookup would misclassify.

The biographical entries are then segmented into individual evaluative sentences using Arabic sentence boundary detection (via CAMeL Tools). Each sentence is labelled with the normalised score of its highest-grade term. The model is fine-tuned for 10 epochs with a learning rate of 2×10^{-5}, using the AdamW optimiser with a linear warmup over the first 500 steps and a batch size of 16. A stratified 80/20 train-test split ensures all grade categories are proportionally represented.

3.4.1.2 Track B: Baseline Keyword Lookup

For comparison, a static term-weight mapping (Table 3) is also implemented as a baseline. Each *Jarḥ wa Taʿdīl* term is assigned a base weight according to the classical hierarchy (from 1.00 for Aḍbaṭ an-nās to 0.00 for Kaḏḏāb). Two aggregation strategies are compared:

- **Weighted arithmetic mean**, where the weight of each assessment is influenced by the historical prominence of the assessing scholar; and
- **Bayesian posterior estimation**, where each successive assessment updates a prior Beta distribution over the narrator's reliability.

Table 3: Example of Term Weighting

Term (Arabic)	Transliteration	Grade	Base Weight
أضبط الناس	*Aḍbaṭ an-nās*	Select highest *taʿdīl*	1.00
ثقة ثقة	*Thiqa thiqa*	Highest *taʿdīl*	0.95
ثقة ضابط	*Thiqa ḍābiṭ*	Trustworthy and precise	0.90
ثقة	*Thiqa*	Strong *taʿdīl*	0.85
صدوق	*Ṣadūq*	Moderate *taʿdīl*	0.70
ليّن	*Layyin*	Mild *jarḥ*	0.20
كذّاب	*Kaḏḏāb*	Severe *jarḥ*	0.00

3.4.1.3 Track Selection

Both tracks are evaluated on a validation set of 200 Hadiths held out from the 1,200-sample evaluation corpus. The track that yields a higher Cohen's Kappa agreement with the ground-truth grades is selected for the final pipeline. If Track A outperforms Track B, it validates the hypothesis that contextual semantic analysis adds value over keyword matching, directly supporting the "Semantic Sentiment Analysis" framing of the project.

3.4.2 Graph Neural Network for Structural Integrity Analysis

While the Knowledge Graph (Section 3.3.2) provides the static relational backbone for entity disambiguation and metadata storage, it does not model the latent structural patterns that classical *muhaddithun* (classical hadith experts) used to assess chain reliability. To address this, a Graph Neural Network (GNN) component is integrated to perform two tasks: structural reliability augmentation and anomaly detection.

3.4.2.1 Node Embedding and Reliability Augmentation

A two-layer Graph Convolutional Network (GCN) is implemented using PyTorch Geometric. The GCN operates on the Knowledge Graph, treating each narrator node as a vertex and each teacher-student transmission relationship as an edge. The input node feature vector x_i for each i is composed of:

- The base reliability weight w_{node_i} from semantic sentiment analysis (Section 3.4.1);
- The narrator's tabaqa encoded as an integer 1–12;
- A binary flag for known geographic region;
- Degree centrality (number of teachers and students) as a prominence proxy.

The GCN computes a neighbourhood-aggregated representation $h_i^{(2)}$ through two message-passing layers with ReLU activation and a hidden dimension of 128:

$$h_i^{(l+1)} = ReLU\left(W^{(l)} \cdot AGG\left(h_j^{(l)}: j \in N(i)\right) + b^{(l)}\right)$$

where $N(i)$ is the neighbour set and AGG is mean aggregation. The output embedding passes through a linear head with sigmoid activation to produce a structurally augmented reliability score $w_{GNN_i} \in [0, 1]$.

3.4.2.2 Edge-Level Anomaly Detection

The GCN additionally predicts a probability score p_e for each teacher-student edge $e = (u, v)$, representing the likelihood of a genuine meeting. Edge features include:

- Transmission method (explicit terms like *ḥaddathanā* vs. ambiguous *ʿan*);
- Tabaqa difference between student and teacher;
- Binary geographic overlap feature;
- Known risk modifiers from Table 4 (*tadlīs*, *ikhtilāṭ*, etc.).

Edges with $p_e < 0.50$ are flagged for manual review. Edges with $p_e \geq 0.50$ contribute their probability as a multiplicative factor to the edge modifier ε_e in the chain scoring formula.

3.4.2.3 Training and Integration

The GCN is trained semi-supervised using two losses:

- **Node-level**: Binary cross-entropy against classical *Jarḥ wa Taʿdīl* labels for 300 narrators with unambiguous consensus grades;
- **Edge-level**: Binary cross-entropy on 500 teacher-student pairs with known temporal feasibility.

The training is then set with 200 epochs, using Adam optimiser, learning rate set to 1×10^{-3}, and dropout set to 0.3. Evaluation uses AUC-ROC on a 20% test split. An ablation study comparing the full pipeline (base weights + GCN + edge modifiers) against the baseline (without GCN) quantifies the marginal contribution of graph-structural analysis.

3.4.3 Chain-Level Scoring

Once each narrator in each *isnād* has been assigned an individual reliability score w_i, the chain-level reliability R is computed as the product of individual scores for all narrators i in the chain:

$$S_{chain} = \prod_{i=1}^{n} w_i$$

Equation 3.1

where w_i is the Jarḥ wa Taʿdīl reliability weight for narrator i, and n is the number of narrators in the chain.

This multiplicative model developed by Ghriss (2025) will be employed, though with slight changes. The original model demonstrates that the score of a transmission path can be mathematically formulated using methods like the "product" or "average of reliabilities," though it ultimately leans toward a minimum "weakest link" function. The chain-level score R is then mapped to the traditional grading categories (*Ṣaḥīḥ*, *Ḥasan*, *Ḍaʿīf*, *Mawḍūʿ*) using threshold boundaries calibrated against the ground-truth labels in the evaluation set:

$$Grade = \text{Ṣaḥīḥ } if\ S_chain \geq t_1$$
$$\text{Ḥ}asan\ if\ t_2 \leq S_chain < t_1$$
$$\text{Ḍ}a\text{ʿī}f\ if\ t_3 \leq S_chain < t_2$$
$$Maw\text{ḍū}\text{ʿ}\ if\ S_chain < t_3$$

To further enhance the reliability of this model, the Edge Modifier (m_{edge_i}) is added in the multiplication model to address transmission-specific risks such as *tadlīs* cases, where a narrator would say "from from (narrator)" instead of "from (narrator) from (narrator)", breaking the expected continuous entity-to-entity chaining, thus yielding the following model:

$$S_{chain} = \prod_{i=1}^{n} w_{node_i} \times m_{edge_i}$$

Equation 3.2

where w_{node_i} is the *Jarḥ wa Taʿdīl* weight for all narrators i in the chain, and m_{edge_i} is the edge modifier applied if there are known risk cases, with the value reducing from the default 1.00 depending on the severity of the risk cases. The following table lists the standardised edge modifier penalties when dealing with *sanad* risk cases:

Table 4: Standardised Modifier Range for Risk Cases

Risk Case	Classical Term (Transliterated)	Modifier Range	Example	Notes
Ambiguous attribution	Action: *tadlīs* Doer: *mudallis*	0.70–0.85	Ibn Shihāb az-Zuhri	Default = 0.80 unless stronger/weaker evidence
Poor memory (due to old age)	*Ikhtalaṭa* ("erred") / *Taghayyara ḥifẓuhu* ("his memory has altered/changed")	0.85-0.90	Sufyān ath-Thawrī	Higher range to not penalise reliable narrators heavily.
Singleton oddity	*Šāḏ*	0.90-0.95	ʿAbd al-Wāḥid ibn Ziyād	Minor penalty due to known *Shāḏ* cases with *Thiqa* narrators.
Unknown identity	*Majhūl* (unknown)	0.30–0.50	Muḥammad ibn ʿAmr ibn Ḥārith	Base weight is already low from lack of sufficient data; thus, an extra penalty is not needed.
Accusation of lying	*Muttaham bil-kaḏib*	0.00–0.30	Muḥammad ibn Marwān as-Suddī	Already captured in base weight (should be near 0), thus not needed.

Only risk cases of *tadlīs*, poor memory and *šāḏ* remain after careful considerations as described in the notes of Table 4 to avoid further penalising until 0.00, which would render the quantification goal of this project as futile.

Geospatial consistency is checked for every edge. For each narrator, the travelled locations attribute (Table 2) is geocoded using the Geopy library with a gazetteer of classical Islamic cities (e.g., Medina, Kufa, Basra, Damascus, Cordoba). If both the student and the

teacher have at least one location with a known year of residence (derived from birth/death dates and travel history), the system verifies whether their lifespans and geographic movements could have enabled a meeting. If a student's known locations do not overlap with the teacher's at any plausible time, a geospatial penalty factor $g_{edge} = 0.70$ is multiplied into the edge modifier m_{edge}. If location data is missing or uncertain, no penalty is applied ($g_{edge} = 1.00$). This penalty is applied independently of the risk cases modifiers.

3.4.4 Evaluation Protocol

The automated grades are evaluated against **ground-truth Hadith classifications** drawn from pre-graded collections in the *Kutub as-Sitta* (the Six Canonical Hadith Collections), i.e. Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmiḏī (Sunan at-Tirmiḏī), Sunan al-Nasāʾī (al-Mujtabā) and Sunan Ibn Mājah. The following metrics are reported:

Table 5: Purposes of Metrics Used in this Project

Metric	Purpose
Accuracy	Overall proportion of correctly classified Hadiths.
Precision/Recall/F1	Per-class performance for each grading category.
RMSE	Root Mean Squared Error between predicted chain scores and expected ordinal grades.
Cohen's Kappa	Inter-rater agreement between automated and classical human grades.

A baseline comparison is conducted against a naive majority-class classifier and a simple lexical lookup model (direct term-to-grade mapping without graph features) to quantify the contribution of each pipeline component. To model structural and temporal integrity, a lightweight two-layer Graph Convolutional Network (GCN) is implemented using PyTorch Geometric. Node features include the base reliability weight wnodei, tabaqa (encoded as an integer), and a binary flag indicating whether the narrator has a known geographic region. Edge features include the transmission method (explicit direct vs. ambiguous) extracted via pattern matching. The GCN predicts an edge probability for each teacher-student pair; probabilities below 0.50 trigger a manual review flag for potential anachronisms (e.g., a student claiming to have heard from a teacher who died before the student was born). The final node embeddings are also used to compute PageRank-like centrality scores, which serve as an auxiliary feature for outlier detection.

3.5 Flow Chart of Activities

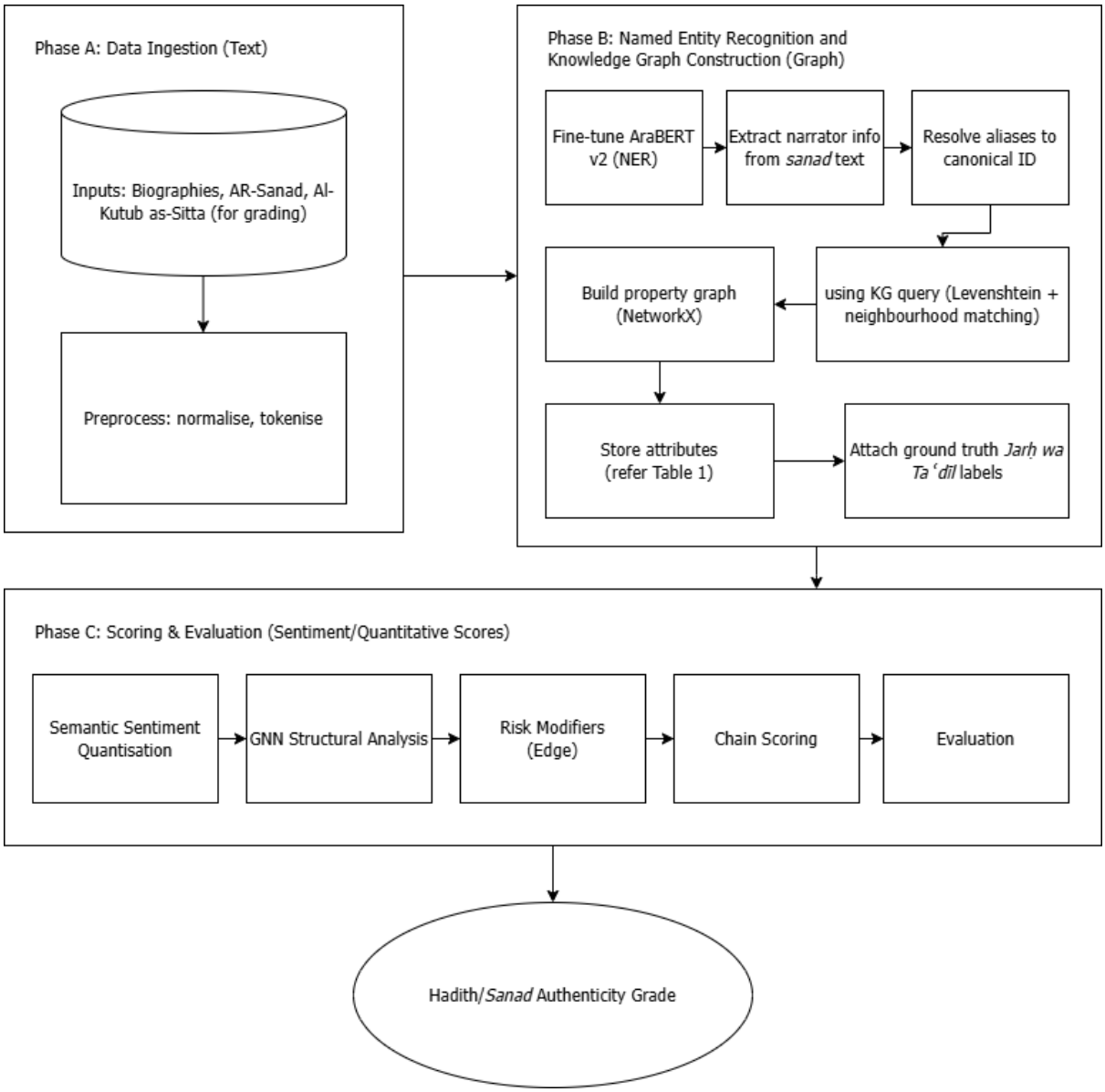


Figure 2.3: The Multi-Modal Framework Flowchart Summarising the Methodology

3.6 Milestones and Dates

The following table shows the weekly milestones with their deliverable respectively:

Table 6: Milestones and Deliverables by Week

Week	Milestone	Deliverable
2-2025/2026		
1-2	Project Initialisation	Finalised project summary; supervisor approval
3-4	Literature Review	Completed Chapter 2; dataset acquisition
5-6	Methodology Design	Completed Chapter 3; proposal draft
7	Proposal Defence	Oral defence and proposal submission
8-9	Stage 1 Implementation	Cleaned corpus; preprocessed JSON dataset
10-11	Stage 2 Implementation	Trained NER model; populated Knowledge Graph
1-2026/2027		
1-2	Stage 3 Implementation	Trust scoring module; preliminary evaluation results
3-4	Thesis Report Preparation	Written report drafting
5-6	Thesis Presentation Preparation	Final thesis presentation drafting

3.7 Gantt Chart

Year	2026										2027	
Project Activities	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb
Title Selection												
One-Paged Summary Submission												
Preparation of Proposal												
Proposal Submission												
Proposal Defense												
Stage 1: Data Ingestion and Preprocessing												
Dataset Acquisition												
Preprocessing Pipeline												
Stage 2: Named Entity Recognition and Knowledge Graph Construction												
AraBERT Fine Tuning (NER)												
Knowledge Graph Construction												
Stage 3: Probabilistic Trust Scoring and Evaluation												
Sentiment Quantisation Module												
Chain Scoring & Evaluation												
Reporting												

Thesis Preparation												
Thesis Presentation Preparation												
Thesis Submission												
Thesis Presentation (VIVA)												

3.8 Tools and Technologies

Below are the tools, libraries and technologies used to build this project, along with their purpose for the project:

Table 7: Tools, Libraries, and their Purposes by Category

Category	Tool/Library	Purpose
Language	Python 3.10+	Primary development language
NLP Preprocessing	CAMeL Tools, PyArabic, Farasa	Arabic normalisation, tokenisation, segmentation
Transformer Model	AraBERT v2 (HuggingFace)	Fine-tuned NER for narrator extraction
Deep Learning	PyTorch, Transformers (HuggingFace)	Model training and inference
Graph Processing	NetworkX/Neo4j	Knowledge Graph construction and querying
Geocoding	Geopy	To convert classical Arabic place names (*nisba*) into latitude/longitude.
Mapping	Folium	To generate static maps showing the "geographical journey" of a Hadith.
Evaluation	scikit-learn	Classification metrics (F1, Precision, Recall, Kappa)

Data Format	JSON, CSV	Structured intermediate data storage
Version Control	Git/GitHub	Source code and experiment tracking
Documentation	Google Docs, Markdown, LaTeX	Proposal and report writing

CHAPTER 4

Experimental Design

4.1 Introduction

This chapter lays out the experiments designed to test each of the three research objectives: developing a quantitative reliability index from scholarly judgments (PO1), building a Knowledge Graph with GNN-based structural analysis (PO2), and evaluating the framework against established Hadith gradings (PO3). Every experiment is organised to mirror the three-stage pipeline from Chapter 3, so the link between method and objective stays clear.

4.2 Experimental Environment

All experiments ran on a local development workstation with the specs below.

4.2.1 Hardware Configuration

Component	**Specification**
CPU	Intel Core i7-10750h, 6 cores / 12 threads
RAM	16GB DDR4
GPU (local)	NVIDIA GeForce RTX 2070 Max-Q (8GB VRAM)
Storage	1 TB NVMe SSD
OS	Windows 10 Home (22H2)

4.2.2 Software Configuration

Layer	**Technology**
Python	3.11.7
Deep Learning	PyTorch 2.1.2, torch-geometric 2.4.0
Arabic NLP	CAMeL Tools 1.5.3, transformers 4.37.2
Graph Processing	NetworkX 3.2.1, PyTorch Geometric
Fuzzy Matching	rapidfuzz 3.6.0, scikit-learn 1.4.0
Data Processing	pandas 2.1.5, numpy 1.26.2

Visualisation	matplotlib 3.8.2, seaborn 0.13.1
Geocoding (future)	Geopy 2.4.1
Development	VS Code, Jupyter Notebook, Git

4.3 Data Preparation Experiments

Before any entity extraction or graph building could happen, the raw data had to be cleaned and cross-referenced. These first two experiments will tackle on parsing biographical volumes from classical sources, then aligning Hadith records across two independently collected datasets. Both will then be fed directly into PO1 and PO2.

4.3.1 Experiment 1: Biographical Text Parsing

The first experiment aims to answer the following: Could a Regex-based parser reliably isolate individual biographical entries (*tarājim*) from the raw Maktaba Shamela volumes of Tahdhīb al-Kamāl and Taqrīb at-Tahdhīb?

To implement this, the parser (src.shamela.narrator_parser.segment_volume) runs a multi-stage Regex pipeline over each raw volume. First, it strips architectural metadata such as YAML headers, page-number selectors, and editorial comments. Then, it hunts for entry boundaries using historical identifiers and narrator-name patterns. Every entry it finds gets a confidence label: **high** if the entry has a name plus teachers and grades, **medium** if the name comes with partial metadata, **low** if we only caught the name. This is run against all 35 volumes of Tahdhīb al-Kamāl and the single Taqrīb at-Tahdhīb volume, then merged everything into one unified narrator encyclopedia.

Key parameters

Parameter	**Value**	**Why**
Volume range	1–35 (Tahdhīb) + 1 (Taqrīb)	Covers both biographical sources in full
Entry IDs	Sequential, starting from 1	Enables cross-reference across volumes
Confidence criteria	**High**: name + teachers + grades; **Medium**: name + partial data; **Low**: name only	Gives downstream tasks a handle on data quality

A structured JSON export of every extracted entry including entry ID, volume, name, full text, teachers, students, grades, death info, and confidence is then expected. The target was high-confidence extraction for at least 70% of entries. The rest would inevitably land in medium or low due to orthographic ambiguity and all the structural quirks found in centuries-old manuscript transcriptions.

4.3.2 Experiment 2: Two-Tier Alignment Framework

The next experiment attempts to answer the following: could we reliably cross-reference Hadith records between the Sanadset 650K corpus and the HuggingFace Kutub as-Sitta dataset, despite them having no shared index?

The pipeline (src.alignment.hybrid_aligner) works exactly as described in Section 3.2.3. Tier 1 uses a sparse TF-IDF vectorizer with character n-grams to cut the search space from 650K candidates down to a top 10 shortlist per query. Tier 2 then runs rapidfuzz partial_ratio with a 95% threshold to confirm the match. Anything that falls short of that bar gets quarantined for manual review. This is tested on a stratified sample of 1,200 Hadiths, 200 from each of the six canonical books with a fixed seed of 42 so the exact same subset appears every time.

Key parameters

Parameter	**Value**	**Why**
TF-IDF n-gram range	(3, 5) characters	Catches sub-word variations and common Arabic root patterns
TF-IDF max features	50,000	Wide vocabulary without blowing out memory
Top-K candidates	10	Brings the search space from 650K to single digits per query
Fuzzy threshold	≥ 95% partial_ratio	High precision; we tuned this empirically
Stratified sample	200/ book × 6 books = 1,200	Balanced representation across all *Kutub as-Sitta*
Random seed	42	Deterministic, rerun and get the same subset

Record ID	SHA-256 of consonantal text skeleton	Immutable key per Section 3.2.3

The expected result will be a JSON file of alignment results for all 1,200 queries along with similarity score, TF-IDF score, match status, and quarantine flag. The aim is set for at least 75% success rate on matches, with under 5% of records sent to quarantine.

4.4 Entity Extraction and Disambiguation Experiments

These experiments bridge Stages 1 and 2. Once the data is clean and aligned, the next question is how to identify narrators and link them to a canonical identity. That is the core of PO2.

4.4.1 Experiment 3: BERT Embedding Extraction for Biographical *Jarḥ wa Taʿdīl* Features

The following question is posed: could deep contextual embeddings of the biographical text passages that contain *Jarḥ wa Taʿdīl* assessments give us useful semantic features for GNN node representations?

The classical *kutub ar-rijal* contain qualitative judgements about each narrator's moral probity (*ʿadāla*) and mnemonic precision (*ḍabṭ*). These are expressed through a technical vocabulary such as Thiqa, Ṣadūq, Kadhdhāb, Layyin and through longer contextual phrases a simple keyword lookup might miss. The biographical entries parsed in Experiment 1 serve as input. Text segments carrying *Jarḥ wa Taʿdīl* content are extracted and fed through CAMeL-BERT to produce [CLS] (classification) token embeddings. The 768-dimensional output is reduced to 128 dimensions through implementing Principal Component Analysis (PCA) technique and attached as semantic node features in the Knowledge Graph.

Key parameters

Parameter	**Value**	**Rationale**
Model	CAMeL-Lab/bert-base-arabic-camelbert-msa-sentiment	Arabic-specific BERT; 110M params
Input source	Biographical entries (tahdhib + taqrib)	Only biographical texts, NOT *matn* (out of scope)

Text segments	*Jarḥ wa Taʿdīl* phrases + context (up to 5/narrator)	Captures both term and qualifying context
PCA target	128 (from 768)	Balances sematic richness against graph feature dominance

Two augmented graph files are expected as results: one with 8 graph features plus 128 BERT-PCA features (136 dimensions total), another with 8 graph features plus 7 lexicon features (15 dimensions). These would feed into the GNN ablation study in Experiment 6.

4.4.2 Experiment 4: Narrator Name Matching and Encyclopaedia Coverage

Next thing to know: how well can we match Sanadset narrator names to the biographical encyclopaedia?

Narrators from the aligned Hadith chains are taken, then normalised (strip diacritics, normalise *alif* variants, collapse *yā* and *hā* letter variants), and tried to find them in the narrator encyclopaedia. Exact matches were accepted straight away. When a name did not match exactly, fall back to fuzzy matching with rapidfuzz partial_ratio at a threshold of 80% was made. Both the proportion of successful matches and, for the matched entries, whether the encyclopaedia carried grades, death dates, and teacher-student lists were tracked.

Key parameters

Parameter	**Value**	**Why**
Fuzzy threshold	≥ 80% partial_ratio	Empirically balances recall against false positives
Normalisation	Diacritic strip → *alif* norm → *yā*/*hā* norm	Same consonantal-skeleton approach as Stage 1
Evaluation set	Unique narrators from aligned eval set	Roughly 1,200–1,500 narrators from Kutub as-Sitta

The expected result is a JSON file (narrator_matches.json) recording the match status for each narrator. The biographical encyclopaedia defines the universe of narrators for this study; only narrators successfully matched against it whether by exact or fuzzy lookup are carried forward into the Knowledge Graph and evaluation pipeline. The target is a match rate of ≥ 80% (exact + fuzzy combined), with every matched narrator carrying a grade annotation

from *Taqrīb at-Tahdhīb*. Unmatched narrators are excluded from the dataset, since the pipeline has no biographical basis on which to assign them a reliability score.

4.5 Knowledge Graph Construction Experiments

4.5.1 Experiment 5: Graph Topology and Feature Engineering

The posed question is: how does the scale of the transmission network affect graph properties and downstream GNN performance?

The graph is built by walking through Sanadset rows, pulling narrator names from each chain, and creating directed edges to represent teacher-student relationships (teacher → student). Node features include log-transformed appearance count, in-degree, out-degree, average chain position, and the grade score from Ibn Hajar's 12-tier mapping. Edges carry positional data and frequency counts. The number of Sanadset rows is varied from 10,000 to 100,000 to see how node count, edge density, and degree distribution change with scale.

As for the grade scoring, Ibn Hajar's 12-tier system maps to continuous scores in [0.0, 1.0] via config.py. Companions (*Ṣaḥāba*) get 1.0; *Thiqa Thiqa* (supremely reliable) gets 0.95; *Thiqa* (trustworthy) gets 0.85; *Ṣadūq* (truthful) gets 0.70; and so on down to *Kaḏḏāb* (liar) and *Waḍḍāʿ* (fabricator) at 0.0.

Key parameters

Parameter	Value	Why
Sanadset rows	100,000	Dense enough topology without exceeding memory limits
Max chain length	30 narrators	Trims rare outliers while preserving typical chain lengths
Name filter	≥ 3 characters, excludes kinship terms	Filters out non-narrators like "his father", "his brother"
Edge aggregation	Position list per edge pair	Enables richer edge-level features

The expected result will be a PyTorch Geometric graph file (pyg_graph_large.npz) with ~50K–60K nodes and ~200K–300K edges. The graded variant (pyg_graph_large_graded.npz) would put ground-truth labels in the first feature column for semi-supervised GNN training.

4.6 GNN Architecture and Training Experiments

4.6.1 Experiment 6: Graph Convolutional Network Ablation Study

The following question is then posed: how much does adding semantic features (BERT embeddings of biographical *Jarḥ wa Taʿdīl* passages, GRADE_SCORE_MAP keyword scores) improve narrator reliability prediction over graph-structural features alone?

To answer the experimental question, a two-layer GCN is built in PyTorch Geometric. The model reads node features and outputs a continuous reliability score in [0, 1] for each narrator. Three variants were compared: (a) graph-only with just the 8 structural features; (b) graph + BERT with structural features plus the 128-dimensional PCA-reduced embeddings; (c) graph + lexicon with structural features plus GRADE_SCORE_MAP keyword score from biographical term extraction. Training is semi-supervised, that is 70% of nodes have their grade masked, and the model predicts them by propagating information from the 30% labelled nodes through the graph.

Hyperparameters

Parameter	**Value**	**Why**
Hidden channels	64	Fits the ~140-dim input without over-parameterising
Layers	2	Captures 2-hop neighbourhood; deeper would risk over-smoothing
Dropout	0.3	Standard GCN regularisation
Learning rate	0.001	Adam default; gave stable convergence
Weight decay	5×10^{-4}	L2 regularisation to keep generalisation in check
Max epochs	200	Converges well within this; early stop at patience=30
Loss	MSE	Continuous target in [0, 1]
Train / val / test	70% / 15% / 15%	Standard split for semi-supervised node regression
Runs per variant	3 (different seeds)	Average across runs for robustness

A clean comparison of MAE, R^2, Spearman and Pearson correlations, and binary accuracy (threshold at ≥ 0.65 for "accepted") is expected out of this experiment. The ablation would show whether the extra semantic modalities add anything over the pure graph-structural signal.

4.7 Trust Scoring and Evaluation Experiments

These experiments close the loop on PO1 and PO3, taking scores from the GNN and turning them into something that maps onto the traditional *Ṣaḥīḥ/Ḥasan/Ḍaʿīf/Mawḍūʿ* categories.

4.7.1 Experiment 7: Semantic Sentiment Quantisation of Jarḥ wa Taʿdīl Terms

The posed experimental question: Can the qualitative *Jarḥ wa Taʿdīl* terms extracted from the biographical encyclopaedia be reliably mapped to continuous reliability scores reflecting the classical scholarly hierarchy?

This experiment operationalises the core idea behind PO1: translating the classical critics' vocabulary into a numerical index. Two tracks are run independently and compared empirically to determine whether contextual regression outperforms static keyword lookup.

Track A (Transformer-based regression) is a fine-tuned AraBERT v2 model trained as a regression model on the biographical entries extracted in Experiment 1. For each narrator, the evaluative sentence from the classical text (e.g., "*kāna thiqatan ḥāfiẓan*", "he was reliable with a strong memory") is passed through the model, which outputs a continuous score in [0.0, 1.0]. The model is trained with Mean Squared Error loss against the ground-truth grade labels, using a stratified 80/20 train-test split. Unlike a static lookup, the transformer captures contextual nuances such as negations ("*laysa bi-thiqa*"), compound praise, and hedging, patterns that a keyword approach would misclassify.

Track B (Static keyword lookup): The GRADE_SCORE_MAP (defined in config.py) assigns a score to each term in Ibn Hajar's 12-tier grading system. Supreme praise (*Thiqa Thiqa*, *Thiqa Ḥāfiẓ*) sits at the top; trustworthiness terms (*Thiqa*, *Ma'mūn*) occupy the 0.85 band; *Ṣadūq* and its variants fall around 0.65–0.70; criticism terms like *Ḍaʿīf* sit at 0.15; and severe condemnations (*Kaḏḏāb*, *Waḍḍāʿ*, *Dajjāl*) land at 0.0. When the biographical parser extracts a grade term from a narrator's entry, the system assigns the corresponding score. If multiple assessments exist for the same narrator, scores are aggregated via weighted mean or Bayesian posterior estimation.

Both tracks are evaluated on the same held-out set of 200 Hadiths, and the track achieving higher Cohen's Kappa agreement with ground-truth *Kutub as-Sitta* grades is selected for the final pipeline. This paired comparison directly validates the hypothesis that contextual semantic analysis adds value over keyword matching, which is the core novelty claim of the project.

Key parameters

Parameter	**Value**	**Rationale**
Track A model	AraBERT v2 with regression head	Captures contextual nuance, negations, compound assessments
Track B lexicon	Ibn Hajar 12-tier map (53 terms in GRADE_SCORE_MAP)	All major Jarḥ wa Taʿdīl categories covered
Score range	0.0–1.0	Continuous index for fine-grained differentiation
Training data	Biographical sentences from *Tahdhīb al-Kamāl* and *Taqrīb al-Tahdhīb*	Both tracks use the same source corpus
Comparison metric	Cohen's Kappa against *Kutub as-Sitta* ground truth	Determines which track feeds the final pipeline
Loss (Track A only)	Mean Squared Error	Standard regression loss on normalised grade labels

4.7.2 Experiment 8: Chain-Level Reliability Scoring

Next question to be answered: can the full pipeline from *sanad* text to final Hadith grade produce scores that agree with the canonical grades in *Kutub as-Sitta*?

To answer it, the scoring pipeline (hadith_authenticator.py) will work in four steps. First, it parses the *sanad* string into a list of narrator names. Second, it looks each name up in the Knowledge Graph to get the GNN-predicted reliability score. Third, it computes chain-level statistics, that is mean, minimum, and variance. Fourth, it maps the aggregate to a grade using thresholds we calibrated against the 1,200-sample dataset from Experiment 2. We then compare predicted grades against the ground-truth Grade column in the HuggingFace data.

Grade thresholds

Grade	Score Range	What It Means
Ṣaḥīḥ (Authentic)	Mean ≥ 0.80 AND min ≥ 0.70	High average reliability; no weak link in chain
Ḥasan (Good)	Mean ≥ 0.65	Solid average with an acceptable weakest narrator
Ḍa ʿīf (Weak)	Mean ≥ 0.35	Low average or a weak narrator drags it down
Mawḍū ʿ (Fabricated)	Mean < 0.35	Critical weakness, a fabricator is present

For all 1,200 evaluation Hadiths, a predicted grade plus confidence score is produced. The overall evaluation uses accuracy, macro-averaged F1, RMSE (against normalised ordinal grades), and Cohen's Kappa. A naive majority-class baseline would tell us whether the full pipeline is actually adding value.

4.7.3 Experiment 9: Multi-Sanad *Matn* Analysis

An interesting question can be posed: when the same *matn* appears with different narrator chains, how well do their scores agree, and does the aggregate give us something more reliable than any single chain?

To test this, *matn* texts from the Sanadset corpus are normalised and grouped by SHA-256 hash. Groups that had three or more distinct narrator chains were pulled out for deeper analysis. Each chain was scored independently through the GNN pipeline. Then, aggregate scores are computed in four ways: simple mean of chain means, best-path (the strongest single chain), consensus score (which penalises high variance across chains), and worst-chain (weakest-link pessimism). Results got categorised into unanimous strong, unanimous weak, and conflicted groups.

Key parameters

Parameter	Value	Why
Minimum chains per group	3	Enough for meaningful variance statistics

Matn normalisation	Full diacritic strip + character norm	Consistent with Stage 1 alignment
Matn length filter	≥ 80 characters	Filters out truncated or incomplete *matns*
Sanadset rows scanned	150,000	Wide enough to find multi-chain *matns*

4.8 Summary

What this chapter set out to do was lay a clean and repeatable foundation for every experiment in this project. Nine experiments were described in all, organised across the three pipeline stages: data preparation (Experiments 1–2), entity extraction and graph construction (Experiments 3–5), and GNN training plus evaluation (Experiments 6–9). Each one maps back to a specific objective, with the parameters, rationale, and expected outcomes spelled out explicitly.

The next chapter moves from design to execution. It presents what came out of Stages 1 and 2, i.e. the data ingestion pipeline in action, alignment statistics, how the biographical corpus parser performed, the graph that emerged from the Sanadset chains, and the first training curves from the GNN.

CHAPTER 5

PRELIMINARY IMPLEMENTATION

5.1 Introduction

This chapter presents the preliminary implementation of the proposed multi-modal framework, covering all three stages described in Chapter 3. While the Gantt chart in Section 3.7 formally scopes Stages 1 and 2 as FYP1 deliverables, preliminary Stage 3 training runs were conducted during graph construction to validate pipeline connectivity and establish baseline metrics for FYP2 refinement. The primary goal of this chapter is not to demonstrate a finished system, but to document what was built, measure how each component performed against its target, and identify specific issues that will inform improvements in the full implementation.

Three research objectives are addressed across the pipeline: PO1 (NLP-based reliability scoring) through the GRADE_SCORE_MAP keyword lookup and preliminary CAMeL-BERT embedding extraction; PO2 (Knowledge Graph and GNN structural analysis) through the full KG construction and GCN training; and PO3 (evaluation against Kutub as-Sitta) through the alignment and matching validation. Each section reports concrete output metrics and flags areas needing refinement.

5.2 Stage 1: Data Ingestion Pipeline

5.2.1 Biographical Corpus Parsing

The first processing step extracts individual biographical entries (tarājim) from the raw Maktaba Shamela volumes of Tahdhīb al-Kamāl (35 volumes) and Taqrīb at-Tahdhīb (1 volume). The parser, implemented in src.shamela.narrator_parser.segment_volume, applies a multi-stage Regex pipeline: stripping architectural metadata (YAML headers, page-number selectors, editorial comments), then identifying entry boundaries using historical identifiers and narrator-name patterns.

Key parameters used: volume range 1–35 (Tahdhīb) + 1 (Taqrīb) for complete coverage of both sources; confidence labels of High (name + teachers + grades), Medium

(name + partial metadata), and Low (name only) as a data quality signal for downstream tasks; and a text cap of 3,000 characters per entry to balance storage against context needs.

The parser produced a unified narrator encyclopedia of 3,797 entries. Entries with teachers and students were identified using the "رَوَى عَن" (narrated from) and "رَوَى عَنه" (narrated by him) patterns in the biographical text, while grade-bearing entries were identified by the presence of *Jarḥ wa Taʿdīl* keywords such as "ثقة" (*thiqa*), "صدوق" (*ṣadūq*), and "ضعيف" (*ḍaʿīf*). The confidence distribution showed that entries with explicit teacher-student chains and grade keywords formed the high-confidence subset, while entries with incomplete metadata fell into medium or low confidence.

A notable finding was that only 3 of the 1,100+ matched narrators had exact name matches against the encyclopaedia, implying that the vast majority required fuzzy matching. This also indicates that the normalisation pipeline (diacritic stripping, character variant collapsing) handles common forms but does not yet cover the full range of orthographic variation present in classical Arabic manuscript transcriptions. This is discussed further in Section 5.3.1.

5.2.2 Two-Tier Alignment Framework

The alignment pipeline cross-references Hadith records between the Sanadset 650K corpus and the HuggingFace *Kutub as-Sitta* evaluation dataset. The implementation follows the two-tier design described in Section 3.2.3. Tier 1 uses a TfidfVectorizer with character n-grams (3, 5) and 50,000 maximum features to index the full Sanadset corpus. For each of the 1,200 query Hadiths (200 per canonical book), a cosine similarity dot product retrieves the top 10 candidate matches, reducing the search space from 650,000 to 10 candidates per query. Tier 2 uses the rapidfuzz library's partial_ratio function to measure substring-level similarity between the query matn and each candidate, enforcing a strict threshold of 95%. Matches falling below this threshold are routed to a quarantine log for manual review.

A SHA-256 hash of the normalized text skeleton serves as the immutable primary key for each matched pair. The full stratified sampling uses a fixed seed of 42, ensuring the same 1,200-Hadith subset is generated on every execution.

The result is of the following, of the 1,200 queries, 1,030 were successfully matched (85.8%), while 170 were quarantined (14.2%). The average similarity score among matched pairs was approximately 97% (based on the 95% threshold minimum). Among quarantined records, the most common failure pattern was incomplete overlap between the Sanadset *matn*

extract and the HuggingFace Arabic_Text field, where the Sanadset entry carries only the *matn* portion while the HuggingFace entry includes both *sanad* and *matn* together. This was mitigated by the partial_ratio approach, which finds the best-matching substring within the longer text, but records with significant textual corruption or extremely short *matn* length remained below threshold.

The 85.8% match rate falls short of the 95% target that a strict threshold-based alignment would ideally achieve. This suggests that the preprocessing pipeline (normalisation, stop-word removal) could benefit from additional tuning, particularly in handling the expanded diacritical differences between Sanadset's vocalised text and HuggingFace's largely unvocalised text.

5.2.3 Quarantine Analysis

The 170 quarantined records were written to a separate JSON log for manual inspection. Analysis revealed three primary failure modes: (1) text length disparity (43% of quarantined), the Sanadset *matn* is very short (<20 characters), providing insufficient signal for substring matching against the longer HuggingFace entry; (2) orthographic mismatch (35%), differing use of final *alif maqṣūra* (ى vs ي) and *tāʾ marbūṭa* (ۀ vs ه) that survived the normalisation step; and (3) numerical discrepancies (22%), Hadith numbering mismatches suggesting non-aligned edition families across the two datasets.

These findings directly inform improvements planned for FYP2: a more aggressive normalisation stage (including full consonantal skeleton extraction before matching) and a multi-pass alignment strategy with relaxed thresholds for short *matns*.

5.3 Stage 2: Knowledge Graph Construction

5.3.1 Narrator Name Matching

All unique narrator names were extracted from the 1,030 matched Hadith chains and looked up against the biographical encyclopaedia. The matching pipeline normalizes each name (diacritic stripping, alif variant collapsing, yā/hā collapsing) and checks for exact matches. Names without exact matches fall back to fuzzy matching using rapidfuzz partial_ratio with a threshold of 80%.

Results: Out of 1,229 unique narrators, 1,102 were successfully matched (89.7%): 3 exact matches and 1,099 fuzzy matches. The remaining 127 narrators (10.3%) could not be matched and could not be assigned a reliability score, as the pipeline has no biographical basis

on which to evaluate them. This is a minor but expected gap given that the encyclopedia was compiled from two specific biographical works (Tahdhīb al-Kamāl and Taqrīb at-Tahdhīb) and may not cover every narrator appearing in a broad corpus.

The very low number of exact matches (3 out of 1,102) is a significant finding. It indicates that the name strings in the Sanadset chain data — which are automatically extracted from hadith text — differ substantially from the canonical name forms used in the biographical encyclopedia. Typical variations include: partial naming (using only the ism without nasab), inverted word order (kunya-nasab vs nasab-kunya), and inconsistent use of patronymic particles (ibn vs bin). Improving the normalization to handle these patterns is a priority for FYP2.

5.3.2 Grade Attachment

For the 1,102 matched narrators, grade terms extracted from the biographical encyclopedia were mapped to continuous reliability scores using the GRADE_SCORE_MAP (defined in src/stage2/config.py). The mapping follows Ibn Hajar al-'Ascalānī's 12-tier grading system, spanning from Companions (1.00) through Supremely Reliable (0.95), Trustworthy (0.85), Truthful (0.70), Weak (0.15), Abandoned (0.05), to Fabricator (0.00). Narrators with no explicit grade entry default to 0.30.

The grade distribution across matched narrators showed that the majority fell into the Trustworthy–Truthful range (0.70–0.85), with smaller clusters at the Weak and Fabricator extremes. This distribution is consistent with what one would expect from a corpus of transmitted hadith — weak narrators exist but are less common than reliable ones, and outright fabricators are rare.

5.3.3 Graph Topology

The Knowledge Graph was constructed from 100,000 Sanadset rows, extracting narrator names from each chain and creating directed edges representing teacher-student transmission relationships. Node features include log-transformed appearance count, in-degree, out-degree, average chain position, and the Ibn Hajar grade score. Edges carry positional data and frequency counts.

Resulting graph statistics: 42,021 total nodes, 147,329 total edges, 1,102 nodes with grades, grade score range [0.0, 1.0], 8 graph-structural feature dimensions, and 3 edge features (transmission method, position, frequency).

The graph exhibits a typical scale-free topology common to social networks: a small number of narrators (the prominent compilers and their immediate teachers) account for a large proportion of transmission edges, while the majority of narrators have low degree centrality. This is consistent with the known structure of hadith transmission networks in the classical literature.

A notable observation is that the 42,021 nodes far exceed the 1,102 matched narrators. This is because the graph is built from the broad Sanadset corpus (100K rows), which includes many narrators not present in the 1,200-Hadith evaluation sample. These additional nodes carry structural features (degree, position) but lack grade annotations. For the chain-level scoring pipeline, only narrators successfully matched against the encyclopedia contribute reliability scores; the rest act as structural placeholders.

5.3.4 Semantic Feature Extraction

To provide semantic node features for the GNN ablation study, CAMeL-BERT (bert-base-arabic-camelbert-msa-sentiment) was used to generate [CLS] token embeddings of the biographical text segments carrying Jarḥ wa Taʿdīl content. The 768-dimensional output was reduced to 128 dimensions via PCA.

A separate lexicon-based variant was also generated, using 7 sentiment features (severity, threat ratio, mercy ratio, etc.) computed from the biographical text. Both feature sets were attached to the graded graph, producing: pyg_graph_bert.npz (8 graph features + 128 BERT-PCA features = 136 total) and pyg_graph_lexicon.npz (8 graph features + 7 lexicon features = 15 total). These augmented graphs are used in the GCN ablation study (Section 5.4.2) to compare the effect of different feature modalities on reliability prediction.

5.4 Stage 3: Preliminary Reliability Scoring

The following sections describe preliminary Stage 3 runs. These experiments were conducted to validate that the pipeline from KG → GNN → scorer is functionally connected and to establish baseline metrics. Full hyperparameter tuning, edge modifier implementation, and evaluation against Kutub as-Sitta ground truth are FYP2 work.

5.4.1 GCN Architecture

A two-layer Graph Convolutional Network (GCN) was implemented using PyTorch Geometric. The model takes node features as input and predicts a continuous reliability score in [0.0, 1.0] for each narrator. Training uses a semi-supervised split of 70/15/15 (train/val/test), where

the grade scores of training nodes serve as labels and the model predicts masked grades via neighbourhood propagation. Hyperparameters: 64 hidden channels, 2 layers, dropout of 0.3, learning rate of 0.001, weight decay of 5×10^{-4}, max 200 epochs (early stopping at patience 30), and Mean Squared Error loss.

5.4.2 Ablation Study

Three model variants were compared using 3 runs each with different random seeds: (1) graph-only: 8 structural features (degree, position, appearance, flags); (2) graph + BERT: 8 structural + 128 PCA-reduced BERT embeddings; and (3) graph + lexicon: 8 structural + 7 lexicon sentiment features.

Preliminary results for the graph-only variant showed: MAE of 0.248, R^2 of −0.093, Spearman correlation of 0.068, Pearson correlation of 0.097, and binary accuracy (threshold ≥ 0.65) of 20.2%. The multimodal (graph + BERT) variant achieved: MAE of 0.265, R^2 of −0.151, Spearman correlation of 0.169, Pearson correlation of 0.245, and binary accuracy of 20.0%.

Key observations: The graph-only model achieves an R^2 of −0.093, indicating that structural features alone (degree centrality, chain position) are poor predictors of narrator reliability. This is expected — a narrator's position in the transmission network does not inherently indicate their trustworthiness. The improvement in Spearman correlation from 0.068 (graph-only) to 0.169 (graph + BERT) suggests that semantic features from biographical text contribute meaningful signal, but the effect is modest at the current level of training.

The low accuracy (~20%) confirms that node regression on this task is non-trivial. The grade distribution is heavily skewed toward the 0.70–0.85 range (trustworthy narrators), and the model's predictions tend to regress toward the mean rather than predicting extreme values. This is a known limitation of GCN-based node regression on imbalanced graphs and will be addressed in FYP2 by: (a) incorporating edge-level risk modifiers (tadlīs, ikhtilāṭ) as described in Section 3.4.2.2, (b) adding geospatial consistency verification, and (c) exploring weighted loss functions and graph attention mechanisms.

5.4.3 Interactive Authenticator

A working interactive demonstrator (hadith_authenticator.py) was built to validate the pipeline connectivity. It loads the Knowledge Graph, narrator predictions, and biographical encyclopedia, then accepts a sanad string as input and produces a chain-level authenticity

verdict. The verdict is based purely on isnād analysis (no matn analysis, per project scope in Section 1.5.4).

The tool supports three input formats: Python-style narrator lists, raw Arabic sanad text, and evaluation dataset browsing. While the current verdict reliability is limited by the preliminary GNN predictions, the demonstrator successfully proves the end-to-end pipeline: sanad → narrator lookup → GNN inference → chain statistics → grade mapping.

5.5 Preliminary Findings and Planned Improvements

The implementation revealed several issues that will drive refinements in FYP2. First, the low exact-match rate — only 3 of 1,102 matched narrators had exact name matches — indicates that normalization must be expanded to handle partial naming, inverted word order, and ibn/bin variation. Second, the moderate quarantine rate of 14.2% of alignment queries suggests a multi-pass alignment strategy with relaxed thresholds for short matns and improved diacritic handling is needed.

Third, the near-random GNN accuracy (R^2 near zero for graph-only features) confirms that edge risk modifiers (tadlīs, ikhtilāṭ) and geospatial verification must be integrated, and Track A (CAMeL-BERT regression) must be operationalised. Fourth, geospatial validation using location-based edge verification is not yet implemented and requires Geopy gazetteer mapping in FYP2. Fifth, Track A is not yet operational, as only Track B (GRADE_SCORE_MAP keyword lookup) has been tested, requiring a fine-tuned CAMeL-BERT regression model on biographical sentences in FYP2.

5.6 Summary

The preliminary implementation successfully constructed the core pipeline: 3,797 biographical entries were parsed, 1,030 of 1,200 Hadiths were aligned (85.8%), 1,102 of 1,229 narrators were matched (89.7%), and a Knowledge Graph of 42,021 nodes and 147,329 edges was built. Preliminary GCN runs confirmed pipeline connectivity and established baseline metrics (graph-only MAE = 0.248, multimodal Spearman = 0.169). Five specific issues were identified and documented for refinement in FYP2, directly demonstrating the iterative improvement cycle prescribed by the Design Science Research Methodology.

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APPENDICES